

duplication of efforts and codes," feels Jean Michel Sellier, the main developer of the software.

## Approaches implemented in the software

The paper, Archimedes, the Free Monte Carlo Simulator: A GNU Package for Submicron Semiconductor Devices on nanoHUB, by Jean Michel Sellier and others, describes the approaches implemented in the software.

**The Monte Carlo method.** The particle nature of electrons in the sub-classical level is exploited in this method. This nature can be completely described with the help of two vectors, the position vector and the pseudo-wave vector. In this method, the two vectors are evolved for each time instant and for every particle. The method also considers the drift of electrons on interaction with the electrostatic potential and the scattering caused due to interactions with lattice photons.

Archimedes uses the ensemble Monte Carlo method to simulate and predict the behaviour of devices. In this method, simultaneous calculation of many particles (ensemble) dynamics during a small interval of time is performed till a final time is reached. This method can be used to get both stationary and transient solutions of Boltzmann transport equation applied to a semiconductor device.

**Quantum effective potential model.** It is very important for semiconductor device simulations to mimic the quantum effects primarily arising from the non-zero size of electron-wave packets. Most of the effective potential models

## Graphical user interface: Making the software usage simpler

Those who do not want to know anything about the scripting language, but still want to run simulations, can use the graphical user interface (GUI). The GUI can be run locally (needs to be installed first) or online in a browser (no installation is needed in this case). The interface is developed using the infrastructure called Rapture. We can simply declare parameters associated with our tool by describing Rapture objects in Extensible Markup Language (XML). Rapture reads this description and generates the GUI automatically. Simulations of Archimedes can also be run directly using a web browser in a free cyber infrastructure for scientific tools called nanoHUB.

## nano-archimedes

This is a TCAD, based on the platform implemented for Archimedes, and is used for modelling nanometre-scaled semiconductor devices. The methodology followed in the software is based on the signed particle formulation of quantum mechanics (a brand new formulation of quantum mechanics), which is a generalisation of Wigner Monte Carlo method.

Dynamics of electrons, such as transport in nanometre-scale semiconductor devices (nanodevices), and dynamics of N-body problems in quantum chemistry can be modelled using this software. The code is able to simulate time-dependent, full quantum, multi-dimensional phenomena, such as wave-phase breaking and single electron ballistic transport with open-boundary conditions and electron dynamics in molecular systems, among others.

We can execute the software in both serial and parallel machines. The parallel execution is based on MPI and OpenMP, the two standard libraries for parallel calculations. The code is entirely written in C language and can compile on a huge variety of machines without any particular effort.

have successfully modelled the behaviours, such as drift diffusion, but unfortunately even the most sophisticated ones fail in modelling effects, like barrier tunnelling and/or source-drain tunnelling. The complexity further increases when used along with a very noisy method, such as Monte Carlo. Models implemented in Archimedes include Bohm potential,

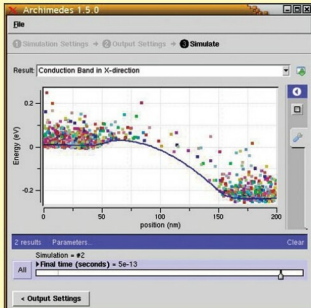


Fig. 2: Simulation of the conduction band in a MESFET

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